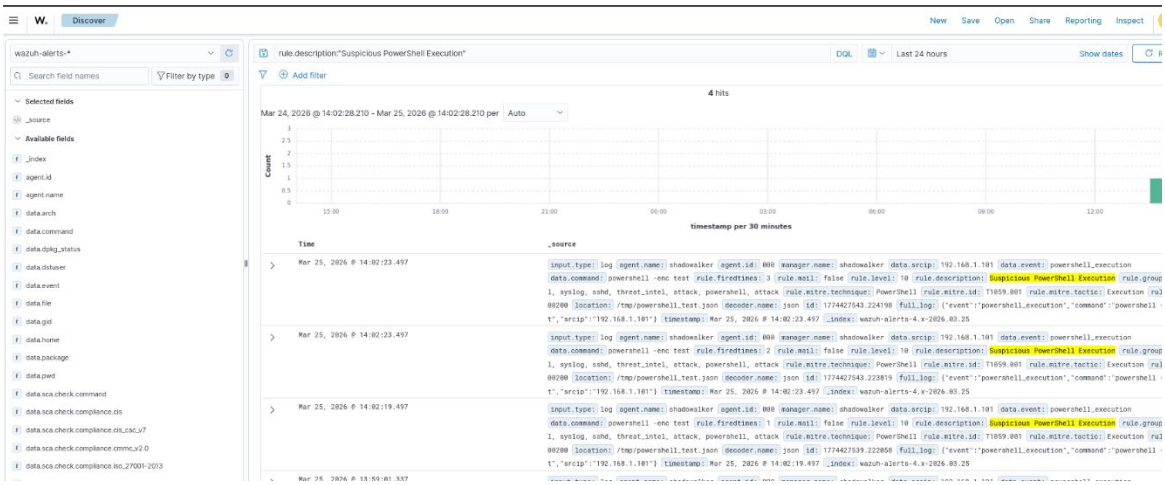
Alert Triage with Automation Report

Avinash Shankar PT
Cyber Security Intern

1. Objective

The objective of this task was to perform alert triage using Wazuh and automate threat validation using VirusTotal integrated with TheHive. The goal was to enhance detection accuracy, reduce manual effort, and improve incident response efficiency within a SOC environment.

2. Alert Information



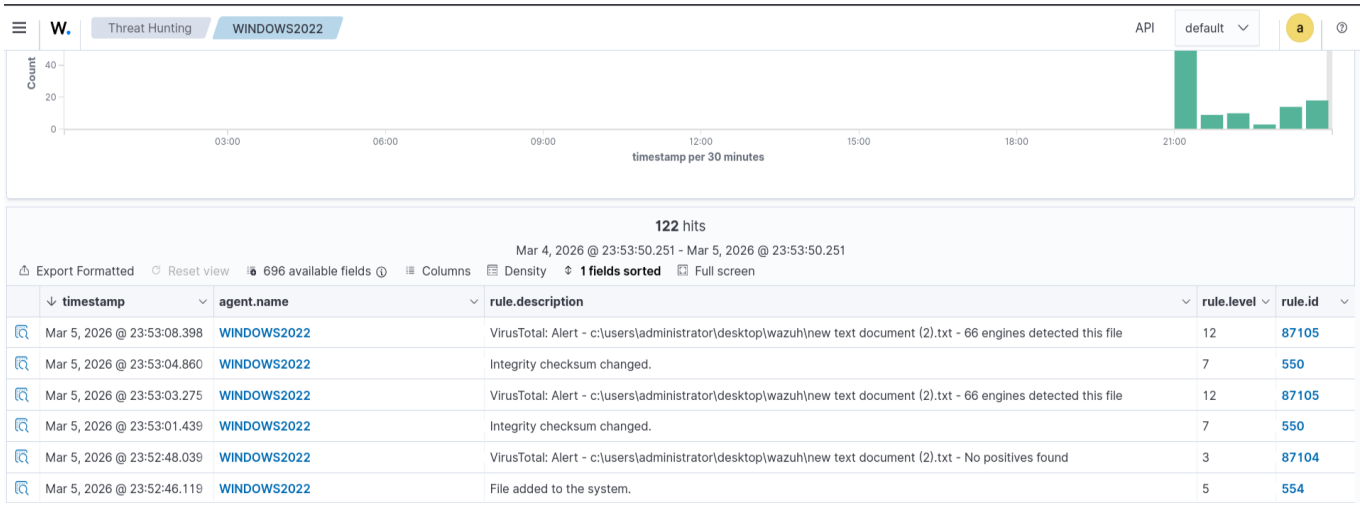
3. Alert Triage Process

The alert was generated in Wazuh, indicating a **suspicious file download activity** originating from the IP address 192.168.1.102. Upon reviewing the alert details, it was classified as **high priority** due to the potential risk of malware infection or unauthorized file execution.

The triage process involved:

- Reviewing alert metadata such as rule description, timestamp, and severity
- Identifying the affected system and user context
- Extracting file-related indicators, specifically the **SHA256 hash**

This initial triage confirmed that the alert required deeper investigation and validation.



timestamp	agent.name	rule.description	rule.level	rule.id
Mar 5, 2026 @ 23:53:08.398	WINDOWS2022	VirusTotal: Alert - c:\users\administrator\desktop\wazuh\new text document (2).txt - 66 engines detected this file	12	87105
Mar 5, 2026 @ 23:53:04.860	WINDOWS2022	Integrity checksum changed.	7	550
Mar 5, 2026 @ 23:53:03.275	WINDOWS2022	VirusTotal: Alert - c:\users\administrator\desktop\wazuh\new text document (2).txt - 66 engines detected this file	12	87105
Mar 5, 2026 @ 23:53:01.439	WINDOWS2022	Integrity checksum changed.	7	550
Mar 5, 2026 @ 23:52:48.039	WINDOWS2022	VirusTotal: Alert - c:\users\administrator\desktop\wazuh\new text document (2).txt - No positives found	3	87104
Mar 5, 2026 @ 23:52:46.119	WINDOWS2022	File added to the system.	5	554

4. Indicator Extraction and Analysis

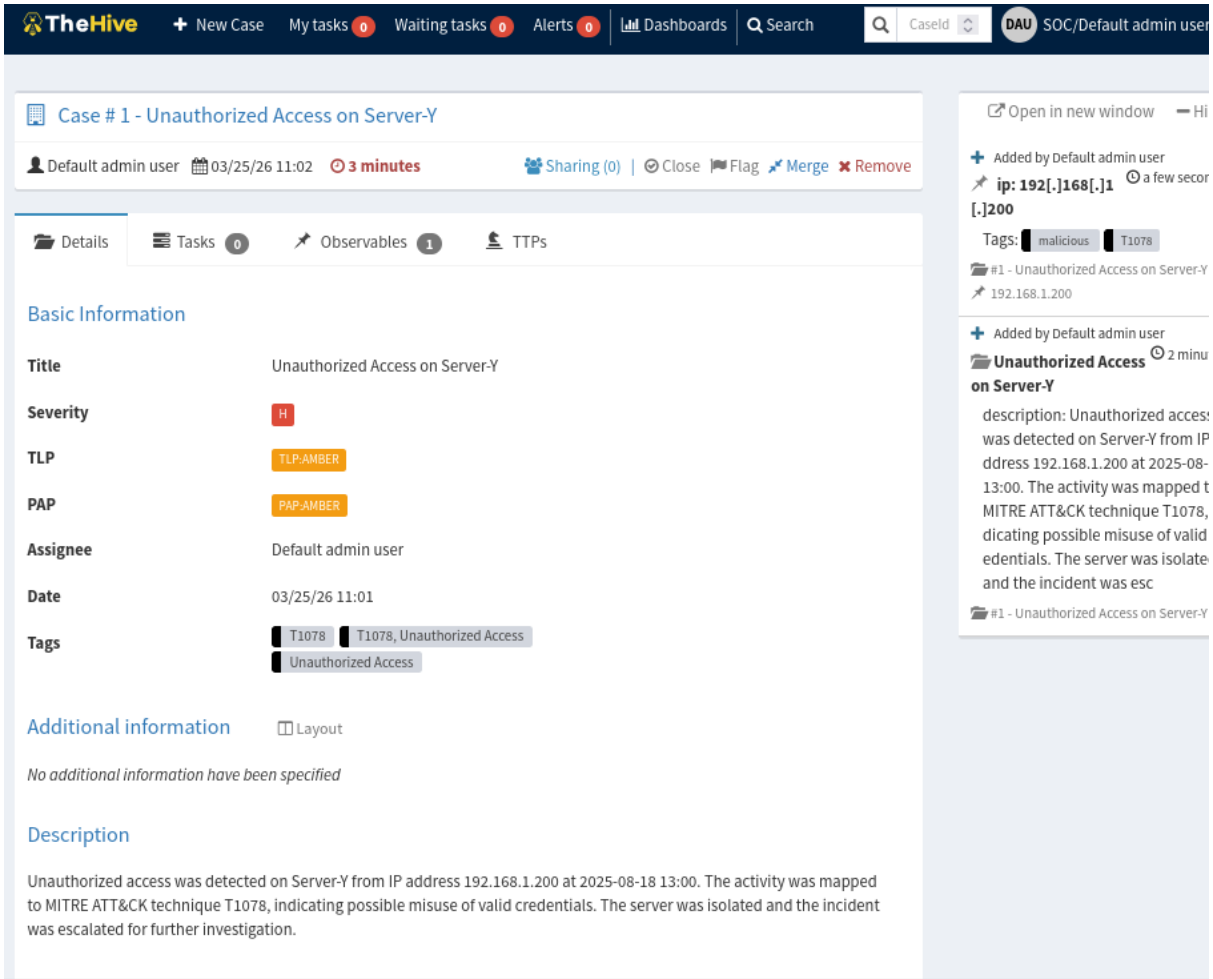
The suspicious file associated with the alert was analyzed by extracting its cryptographic hash (SHA256). This hash uniquely identifies the file and enables reputation checks across threat intelligence platforms.
If the hash was not directly available in Wazuh logs, it was manually generated using system tools. This ensured that the exact file under investigation could be validated accurately.

5. Automated Threat Validation

To automate the validation process, the extracted file hash was submitted to TheHive as an observable. TheHive was configured with VirusTotal analyzers to automatically query the file reputation.
Steps involved:

- Adding the hash as an observable in TheHive
- Running or auto-triggering the VirusTotal analyzer
- Collecting scan results from multiple antivirus engines

The VirusTotal analysis revealed that the file was flagged as **malicious by several detection engines**, confirming the presence of a real threat.



6. Response and Remediation

Based on the validation results, the following response actions were recommended:

- Escalation of the alert to incident response
- Blocking of the source IP address (192.168.1.102)
- Isolation of the affected endpoint
- Removal or quarantine of the malicious file
- Updating detection rules to prevent similar threats

Automation ensured that these steps could be initiated quickly with minimal manual intervention.

7. Automation Benefits

The integration of TheHive with VirusTotal provided several advantages:

- Faster threat validation
- Reduced analyst workload
- Improved accuracy in identifying malicious files
- Consistent and repeatable triage process
- Enhanced overall SOC efficiency

8. Result Summary

The suspicious file download alert was triaged using Wazuh, and the file hash was extracted for analysis. TheHive automated validation through VirusTotal integration, identifying the file as malicious. This process improved investigation speed, reduced manual effort, and ensured accurate threat detection, strengthening the overall efficiency of the SOC workflow.

9. Conclusion

This task demonstrated the importance of combining SIEM, SOAR, and threat intelligence platforms for effective alert triage. By leveraging automation, the SOC team can quickly validate threats, respond efficiently, and minimize potential damage. The workflow highlights a scalable and efficient approach to modern security operations.